The Hidden Tax of Inflation: Fiscal Drag in Italy and Across Europe

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Abstract

This thesis examines fiscal drag: the hidden rise in the effective tax burden that occurs when inflation meets a progressive personal income tax whose parameters are not automatically indexed. When brackets, deductions, and credits stay fixed in nominal terms, a rise in nominal income can push taxpayers into a higher bracket or erode the real value of their allowances, even though their purchasing power has not changed. Taxation rises with no explicit legislative decision behind it, and the consequences reach equity, disposable income, and the transparency of fiscal policy. The study focuses on the Italian personal income tax (IRPEF) during the post-pandemic inflation, across 2020–2024 and the reforms adopted between 2022 and 2024. The first chapter builds the theoretical framework: how progressive taxation works, why average and marginal rates diverge, what brackets actually do, and how nominal income comes apart from real income. From there it defines fiscal drag and shows that a non-indexed schedule can raise the average tax rate even when real income has not grown at all. The second chapter brings this framework to Italian data. Using administrative records from personal income tax returns published by the Department of Finance, the thesis constructs a counterfactual in which the 2020 IRPEF brackets are adjusted for cumulative inflation. Fiscal drag is then measured as the gap between the tax actually paid under the observed rules and the tax that would have been due under indexed brackets. The design has one clear purpose: to separate what inflation did from what discretionary reform did. The results are less uniform than the standard account suggests. Fiscal drag in Italy is real and measurable, but it does not climb steadily year after year. In 2021 and 2023 the effect is positive, as nominal income growth runs up against fixed tax parameters. In 2022 and 2024 the picture reverses: discretionary reforms to the IRPEF schedule absorb part of the bracket creep and lower the effective burden. The weight also falls unevenly. Middle incomes are the most exposed, because they sit closest to the bracket thresholds; low incomes stay partly below the tax base; and once a taxpayer is already in the top bracket, additional drag has little effect. The third chapter reads the Italian case against two others, Germany and Sweden. Germany keeps bracket creep in check

through regular legislative adjustments; Sweden relies on a more rule-based annual indexation, though not without exceptions. The contrast is telling: in the absence of a systematic rule, correcting fiscal drag in Italy is left to discretionary political choice. The thesis argues that indexing the schedule to a domestic consumer price index could make the system more transparent, more predictable, and fairer in horizontal terms. It also argues against treating full indexation as the only remedy. Partial or threshold-based indexation could keep part of the tax system's stabilising role intact while limiting the unintended redistributive effects of inflation.

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Introduction

"Inflation is the one form of taxation that can be imposed without legislation." Milton Friedman's remark captures a mechanism that still operates, quietly, inside modern tax systems. When prices rise, a progressive income tax can take a larger share of a household's income without any vote in parliament and without any real gain for the taxpayer. This is fiscal drag, and it is the subject of this thesis.

The mechanism can be stated simply. In a progressive system, income is taxed at rates that rise with the size of the tax base. If inflation lifts nominal incomes while the brackets, deductions, and credits stay fixed, part of taxpayers' income is pushed into a higher band, and the effective average tax rate rises even though purchasing power has not changed. The increase is real, but it follows from no policy decision. It is, in Friedman's sense, a tax imposed without legislation. The phenomenon is not marginal in scope. Calvo and Sánchez de la Cruz (2024) report that, of 160 economies worldwide, 131 have no automatic mechanism to adjust tax thresholds to inflation, and they estimate that fiscal drag alone raised the EU tax wedge by 0.84 percentage points between 2019 and 2022. Italy belongs to the larger group: the IRPEF has never had a general rule for indexing its parameters to prices.

The post-pandemic years gave the question new urgency. After two decades of low inflation, prices in Italy rose sharply in 2022 and 2023, the steepest increase in four decades. Nominal incomes increased, while real incomes did not, and a tax system with fixed thresholds was left to operate in conditions it had not faced for a generation. This is the setting in which the thesis asks whether, and by how much, fiscal drag raised the Italian tax burden, who bore it, and what could be done to contain it.

The work proceeds in three steps. The first chapter sets out the theory: the structure of progressive taxation, the distinction between nominal and real income, and the mechanism through which inflation and bracket creep raise the effective tax rate, together with the distributive and macroeconomic effects that follow. The second chapter turns to Italy. Using the administrative microdata on personal income tax returns published by the Department of Finance for the tax years 2020 to 2024, it builds an inflation-indexed counterfactual: for each income cell, it compares the tax actually paid with the tax that would have been due had the 2020 brackets been adjusted for inflation. The difference provides an estimate of fiscal drag. The third chapter places Italy in a European context, comparing it with Germany, which corrects bracket creep through frequent legislative adjustment, and Sweden, which indexes its thresholds by rule. It then returns to Italy with a policy proposal, an indexation mechanism tied to the consumer price index, and tests it on the same data used in the empirical chapter.

Three results emerge. First, fiscal drag is present and measurable, but it does not move in one direction. In the years without reform, 2021 and 2023, the data show a positive fiscal drag that grows with inflation, which is essentially bracket creep; in the reform years, 2022 and 2024, discretionary rate cuts more than offset the drag and lowered the effective rate. Italy, in other words, does correct for inflation, but through irregular reforms timed by the budget cycle rather than by any rule tied to prices. Second, the burden falls unevenly: the middle of the income distribution is the most exposed, since these taxpayers sit above the no-tax area but below the top rate, where a rise in nominal income crosses a threshold most easily. Third, the comparison and the policy simulation point to the same conclusion. A rule that indexed the IRPEF parameters to inflation, even a partial one triggered only above a set threshold, would have neutralised most of the fiscal drag of the period through two adjustments, and would have done so transparently, whereas the actual system left the largest single-year erosion, that of 2023, uncorrected. The central argument of the thesis follows from these findings: the difficulty in the Italian case is not the level of taxation, but the absence of a predictable rule for keeping the tax system aligned with prices.

While the microsimulation approach to bracket creep is well established, this thesis is, to my knowledge, one of the few to apply it to the Italian IRPEF over 2020–2024 using MEF administrative cell data, to decompose measured drag into pure creep and discretionary reform, and to test a triggered indexation rule against full indexation on the same data.

1. Progressive Taxation, Inflation and Fiscal Drag: A Theoretical Framework

1.1. The Progressive Tax System

The progressive tax system is best understood against the broader function of taxation. Taxes are the principal source of government revenue, financing public services, infrastructure, transfers, and social protection. Tax systems, however, do more than raise revenue: they also determine the distribution of income and wealth across the population, a question central to public economics. The principle most commonly invoked to justify taxation is the ability-to-pay principle, according to which individuals with greater financial capacity should bear a higher tax burden than those with less. The principle is intuitive but difficult to implement, since determining an individual's appropriate tax liability requires normative judgments about fairness that are seldom uncontested.

Two criteria are generally derived from this principle. Horizontal equity requires that taxpayers with equal ability to pay be subject to equal tax treatment. Vertical equity requires that taxpayers with greater ability to pay bear a higher tax burden. The progressive tax system is, in effect, an application of vertical equity. Formally, a tax is progressive if the average tax rate, defined as total tax liability divided by taxable income, increases with income. Under such a system, higher-income taxpayers pay a larger absolute amount of tax and a larger share of their income. This stands in contrast to proportional, or flat, taxation, under which all taxpayers are subject to the same rate irrespective of income, and to regressive taxation, under which lower-income households bear a comparatively higher tax burden, generally because a larger share of their income is allocated to taxable consumption.

1.1.1. The Concept of Tax Progressivity

What makes a tax progressive is not the size of the payment, but the rate. A taxpayer earning more does not simply pay a larger sum; they pay a larger share of their income. Under a proportional tax, higher earners also pay more in absolute terms, which can create the impression of a heavier burden, but their income share stays flat. Progressivity requires that share to grow.

This reading of progressivity through the average tax rate follows Rosen, Gayer and Civan, who measure it as the ratio of taxes paid to income. The table below summarises how that rate behaves under the three possible tax structures:

Table 1. Classification of Taxes

Type of Tax	Behaviour of the average tax rate	Meaning
Proportional Tax	Constant	The same share of income is paid in taxes
Progressive Tax	Increasing	Higher-income taxpayers pay a higher share of income
Regressive Tax	Decreasing	Higher-income taxpayers pay a lower share of income

To understand how the progressive tax system works, let us look at a simple example. Consider two taxpayers: Taxpayer A and Taxpayer B

Table 2. Example of Progressive Taxation Based on Average Tax Rates

Taxpayer	Taxable Income	Tax paid	Average tax rate
A	€ 20,000	€ 4,000	20%
B	€ 50,000	€ 15,000	30%

The average tax rate for taxpayer A stands at 20%, against 30% for taxpayer B. Taxpayer B pays more in absolute terms and also faces a higher average tax rate. This is the defining feature of a progressive tax system.

Progressivity is not simply a technical property of a tax schedule. Behind it lies a distributive judgement: those with greater economic capacity should contribute a larger fraction of their income, not just a larger sum.

1.1.2. *Average and Marginal Tax Rates*

Two concepts are central to progressive taxation. They measure different things, and mixing them up leads to real confusion. The average tax rate is the simpler of the two. Divide total taxes paid by total income:

$$t_{average} = \frac{T(y)}{Y}$$

Where $t_{average}$ is the average tax rate, $T(Y)$ is total tax liability and Y is the taxable income. Take a taxpayer earning €40,000 who pays €10,000 in taxes.

$$t_{average} = \frac{10,000€}{40,000€} = 25\%$$

That is the average rate, one euro out of every four is therefore handed over to the state. Now suppose the same person receives an extra €1,000. Their tax bill goes up by €350. That additional €350 on that additional €1,000 gives a marginal rate of 35%, a figure entirely independent of the 25% calculated before:

$$t_{mg} = \frac{\Delta T(Y)}{\Delta Y}$$

Where t_{mg} is the marginal tax rate, $T(Y)$ is total tax liability and ΔY is the change in taxable income.

Rosen, Gayer and Civan say, in their book *Public Finance*, that the marginal rate is the proportion of the last unit of income taxed by the government. It says nothing about the overall burden. A taxpayer can face a 35% marginal rate and still have an average rate well below that.

In a progressive system, this gap is the norm. The marginal rate sits above the average rate and rises faster. Earning more pulls both upward, but not in equal measure. This asymmetry is not incidental; it is precisely what progressivity is designed to produce.

1.1.3. *Progressivity and Tax Brackets*

Progressivity, understood as the principle that taxpayers with a larger tax base should bear a relatively greater tax burden, can be achieved through several mechanisms. One of them is by tax brackets, whereby taxable income is divided into different portions and each portion is taxed at a specific marginal rate. Tax brackets provide the most common system used to reach progressivity, and this structure remains central to the analysis of progressive income taxation in public finance (Bosi, 2023).

The tax base or taxable income, the income that, after the relevant deductions and exemptions, is subject to taxation, is divided into brackets. Higher statutory tax rates are applied only to the portion of income falling within each bracket.

To see how a bracket tax system actually works, an intuitive example is provided:

Table 3. Simplified Progressive Tax Schedule

Brackets	**Statutory tax rate**
0–1,000	5%
1,001–2,000	6%
Above 2,000	7%

Source: Bosi, P. (ed.) (2023) *Corso di scienza delle finanze*. 9th ed. Bologna: Il Mulino, ch. 3, sections 2.4–4.3.

Now imagine we have three different taxpayers: Taxpayer A, Taxpayer B and Taxpayer C, with different income levels. Taxpayer A has a tax base of 500, Taxpayer B of 1,500 and C of 2,500. According to the table above the schedule is: 5% on the first 1000, 6% on the next 1,000, 7% on anything above 2,000. Taxpayer A has a tax base of 500 and never leaves the first bracket:

$T(y)$ = 500 (0.05) = 25.

The average and marginal tax rates are both the same since his income is all included in the first bracket.

$t_{average} = T(y)/Y$ = 25/500 = 5% t_{mg} = 5%

Taxpayer B earns 1,500 and crosses into the second bracket by 500

$T(y)$ = 1,000 (0.05) + 500 (0.06) = 50 + 30 = 80

That extra 500 taxed at 6% is enough to push his average rate above 5% and therefore, we have different values of average and marginal tax rates.

$t_{average} = T(y)/Y$ = 80/1,500 ≈ 5.33% t_{mg} = 6%

Taxpayer C reaches all three brackets with a tax base of 2,500:

$T(y)$ = 1,000 (0.05) + 1,000 (0.06) + 500 (0.07) = 50 + 60 + 35 = 145

His average rate is 5.8%, noticeably higher than that of Taxpayer A, but still a long way from the 7% he pays on his last euro of income:

$t_{average} = T(y)/Y$ = 145/2,500 = 5.8% t_{mg} = 7%

It is evident that when income increases, the marginal rate jumps from one bracket to the next, while the average rate rises more slowly and never reaches the same level. The only exception is Taxpayer A, whose income stays within the first bracket, so his two rates are equal. The graph below shows this clearly.

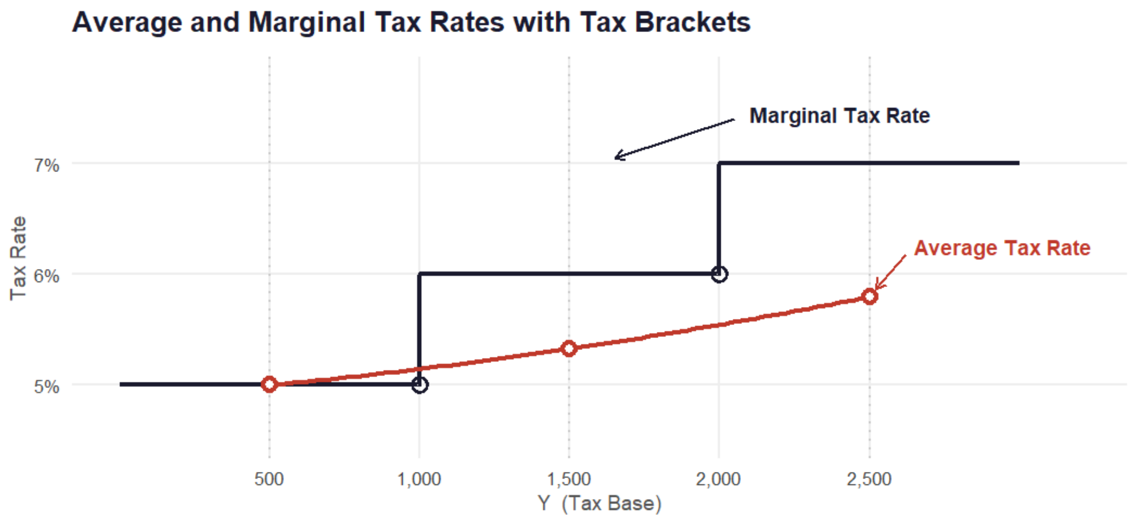

Figure 1. Average and Marginal Tax Rates in a Simplified Progressive Tax Schedule

1.2. Inflation and Income

In this section of the first chapter, we will address the phenomenon of inflation and how it is measured. Inflation is not only the rise in the general price level, but also the erosion and loss of our purchasing power. For this reason, we will first specify the distinction between nominal income and real income and, furthermore, the importance of these two not diverging too much.

1.2.1. Nominal Income versus Real Income

One of the most important things taught to beginner students in a macroeconomics course is the difference between nominal wage and real wage. With the term *nominal wage* we simply refer to the monetary amount received by a worker, while *real wage* measures the purchasing power of that amount after accounting for changes in prices (Stein, 2012). In other words, nominal income tells us how many euros a taxpayer receives, while real income tells us how many goods and services that income can actually buy. The formula adjusted with inflation is:

$$Y_{t+1}^{real} = \frac{Y_{t+1}^{nominal}}{1+\pi}$$

where Y_{t+1}^{real} is real income, $Y_{t+1}^{nominal}$ is nominal income, and π is the inflation rate at year t. For example, a 6% wage increase is a good thing, but if the general price level also rises by 6%, you earn more while your purchasing power is actually the same.

The most commonly used method by economists to estimate the value of the real wage is through the market basket. They are interested in looking at how many of these goods you can actually purchase given a level of income. In Italy, inflation is measured by ISTAT through consumer price indices, CPI, such as the NIC, the FOI and the HICP.

1.2.2. Consumer Price Indices and the Measurement of Inflation

Inflation, defined by ISTAT as a continuous and generalised increase in the price level of goods and services intended for consumption, is measured directly through the construction of a Consumer Price Index (CPI). This index aims to measure the change in prices of a specific, previously mentioned, basket of goods and services representative of household consumption.

The basic approach consists of taking a representative basket of goods and services purchased by households in two distinct periods, and calculating its cost using current period prices in the numerator and base period prices in the denominator. An index built in this way, on a fixed base-period basket, is a Laspeyres index. The calculation process takes place in two distinct phases: in the first phase we calculate elementary indices by category, since the prices of goods and services do not grow proportionally with inflation, while in the second phase we aggregate these indices using a weighted average, where the weights reflect the relative importance of each individual category. However, the CPI should not be interpreted as a mechanically perfect measure of inflation. Nevertheless, the CPI may be affected by weighting bias when fixed weights do not reflect changes in consumers' spending patterns (Bryan and Cecchetti, 1993).

A national Italian index such as the NIC is appropriate for evaluating the erosion of domestic tax thresholds, whereas the HICP is particularly useful for cross-country comparisons within Europe. This distinction will be further developed in Chapter 3, where the role of different indexation mechanisms is examined in a comparative European perspective.

1.3. The Mechanism of Fiscal Drag

Finally, after discussing some theoretical aspects and essential definitions needed to understand the phenomenon, we can begin to address the central topic of this study: fiscal drag.

Fiscal drag, also referred to in Italian as *drenaggio fiscale*, represents one of the most subtle yet significant phenomena in the field of public economics, as it can increase taxpayers' fiscal burden without any direct political or legislative intervention. The interaction between inflation and personal income taxation was analysed extensively by Tanzi (1980), who showed that inflation can distort both the level of taxation and its distribution among taxpayers. Its relevance becomes particularly evident during periods in which the economy is characterised by high inflation. In such circumstances, as previously discussed, nominal incomes increase and the progressive structure of taxation becomes more burdensome, causing these higher incomes to fall into higher tax brackets.

In this section of the first chapter, the aim is to understand how this mechanism operates. Later, the analysis will focus on the possible solutions to what may be defined as an unfair problem.

1.3.1. Defining Fiscal Drag

Fiscal drag is defined as the increase in the tax burden, more precisely in the effective average tax rate, generated by inflation in the presence of a progressive income tax system. When nominal incomes rise because of inflation, and therefore without any real increase in purchasing power, taxpayers may be pushed into higher income brackets and are consequently subject to a higher marginal tax rate.

Moreover, this phenomenon also occurs when real incomes remain unchanged, meaning that the additional tax burden is not the result of a greater ability to pay, but rather a simple consequence of the nominal growth in income.

1.3.2. Bracket Creep and Non-Indexed Tax Systems

The mechanism through which fiscal drag operates is known as bracket creep, namely the "sliding" of income into higher income brackets. In a progressive tax system, nominal income is taxed at increasing marginal rates whenever predefined thresholds are exceeded. The problem arises when these thresholds are fixed and are not adjusted in line with rising inflation, therefore, even a small inflation-driven increase in nominal income results in a portion of that income being taxed at a higher rate than in the previous period. Two distinct factors contribute to this phenomenon and operate simultaneously. The first concerns the structure of progressive tax brackets. When these thresholds remain fixed and are not indexed, indexation being the mechanism through which brackets, deductions and thresholds are adjusted to inflation, the taxpayer is consequently required to pay more.

The second mechanism concerns tax credits and deductions. In many complex systems, such as the Italian IRPEF system, taxpayers may be entitled to income-related tax credits that reduce gross tax liability, in order to reduce the effective tax burden for specific categories of taxpayers, such as employees, pensioners and households with dependants. These amounts depend on the level of income.

However, since these amounts are not updated in line with inflation, their real value, together with that of income, decreases over time.

The two mechanisms combine and jointly produce an increase in the effective average tax rate, without the legislator having formally modified either the tax rates or the tax structure *(Balladares and García-Miralles, 2024).*

1.3.3. A Numerical Illustration of Fiscal Drag

To move away from the theoretical dimension and illustrate the operation of fiscal drag in concrete terms, consider an employee earning an income of €40,000 in year *t*. In the following year, *t+1*, due to an inflation rate of 5%, the employee's nominal income increases to €40,000 (1 + 0.05) = €42,000, while their real income remains unchanged.

Assuming that the structure of the Italian IRPEF system remains unchanged, the tax liability rises from €9,772 in year *t* to €10,645 in year *t+1*, resulting in an overall increase of €873. Looking more closely, by calculating the average tax rate, namely the percentage of total income absorbed by taxation, the result is 24.4% in year *t* (€9,772 / €40,000) and 25.3% in year *t+1* (€10,645 / €42,000). The increase in the average tax rate, equal to 0.9 percentage points, is the result of two forces operating simultaneously: the larger share of taxable income subject to the 35% marginal tax rate and the reduction in the employee tax credit. More specifically, the €873 increase in tax liability can be decomposed into €700 due to the application of the higher marginal tax rate to the additional €2,000 of nominal income, and €173 due to the reduction in the tax credit.

If the tax system had been indexed to inflation, the average tax rate would have remained unchanged at 24.4% for both years, and the tax liability would have increased only to €9,772 (1.05) = €10,261. In the absence of indexation, the fiscal drag effect is therefore equal to €384 (€10,645 - €10,261), calculated as the difference between the actual tax liability of €10,645 and the indexed tax liability of €10,261.

This is the effective amount that the taxpayer is required to pay to the tax authorities not because they possess a greater ability to pay, but exclusively as a result of inflation.

1.4. The Economic Implications of Fiscal Drag

If the previous section analysed the mechanism through which fiscal drag arises, attention must now shift to its direct economic consequences, which are broader and more profound. Fiscal drag is not merely a technical or accounting phenomenon affecting only the relationship between the taxpayer and the tax authorities; rather, it is also highly relevant to fundamental dimensions of the economy, such as income distribution, private consumption, and macroeconomic stability. The following sections will examine these effects in greater detail.

1.4.1. The Increase in the Effective Tax Burden

The most immediate effect of fiscal drag, as seen previously, is the increase in the effective average tax rate borne by taxpayers, which occurs automatically in the absence of indexation of tax parameters. Sanz Sanz and Arrazola Vacas (2025) distinguish between two main components. The first component, referred to as the static component, emerges as a result of the inflationary erosion of the nominal thresholds of tax brackets, independently of any variation in income. The second, dynamic component is linked to the increase in taxpayers' nominal income in response to inflation. According to the empirical analysis conducted by the two Spanish authors, the static component is clearly dominant, accounting for 94% of total bracket creep.

From a comparative perspective, Calvo and Sánchez de la Cruz (2024) documented that, out of 160 economies worldwide, 131 do not provide for the use of an automatic mechanism to adjust tax thresholds to inflation. Furthermore, they estimate that, over the period from 2019 to 2022, the tax wedge in the EU increased by 0.84 percentage points as a result of fiscal drag alone. The absence of indexation therefore allows states to benefit from higher real revenues without resorting to any form of legislative intervention, thus making this phenomenon a form of implicit taxation that is as effective as it is invisible.

1.4.2. Redistributive Effects across Income Groups

Having understood how states benefit from fiscal drag, the following question must be addressed: does fiscal drag affect all taxpayers in the same way, or does it produce differentiated effects across the income distribution?

Fiscal drag does not affect all taxpayers uniformly; rather, it produces different effects depending on the level of income and on the taxpayer's position within the tax system. Middle- and upper-middle-income earners are the most affected, namely those whose incomes are already subject to a moderate level of taxation and who can more easily be pushed into higher tax brackets with correspondingly higher rates. Very low-income earners, on the other hand, may be less exposed, as they may fall within areas of tax incapacity. Fiscal drag, therefore, is not limited to increasing public revenue for the state, but may also have redistributive relevance.

1.4.3. Disposable Income, Consumption and Automatic Stabilisation

Although the existing literature on fiscal drag focuses primarily on its revenue and distributive effects, the phenomenon can also be linked to disposable income and the consumption associated with it. Indeed, if inflation increases nominal incomes without generating an increase in purchasing power, and the system is not indexed, the tax burden may rise. This reduces disposable income and lowers taxpayers' levels of consumption. Households with lower incomes, or with a lower capacity to save, generally tend to allocate a larger share of their income to consumption, whereas those with greater economic resources can more easily absorb a temporary reduction in net income. From this perspective, fiscal drag may have macroeconomic effects, since lower consumption reduces aggregate demand.

In response to this issue, fiscal drag itself can be interpreted as an automatic stabiliser. The increase in the tax burden occurs without discretionary government intervention, and this may help to contain the growth of consumption. However, during periods of high inflation, this mechanism becomes ambivalent, since it may further compress households' real disposable income.

2. Fiscal Drag in Italy: Descriptive Evidence and Counterfactual Analysis

Having examined the theoretical foundations of fiscal drag, this chapter turns to the Italian case. The analysis here is institutional and descriptive: Italy is a relevant case study precisely because IRPEF has no general mechanism for indexing tax brackets and other income-related parameters to inflation. The chapter first sets out the structure and recent evolution of the IRPEF system, then examines the macroeconomic context of the post-pandemic inflationary period, comparing inflation, household disposable income, and IRPEF revenue. The goal is not a causal estimate of fiscal drag, but an assessment of the conditions under which it can arise, and of whether the macroeconomic trends observed are consistent with the mechanism set out in Chapter 1.

2.1. The structure of the Italian IRPEF

IRPEF is a progressive Italian tax that achieves progressivity through income brackets applied to the income of natural persons, as regulated by the Consolidated Income Tax Act *(TUIR, D.P.R No. 917/1986, Art. 11)*. The calculation mechanism involves the application of increasing marginal tax rates to the different taxable income brackets, so that each rate applies exclusively to the portion of income falling within the relevant bracket. The three-bracket structure was introduced for the 2024 tax year by Legislative Decree No. 216/2023 and was then confirmed from 2025 by Law No. 207/2024.

Table 4. IRPEF for the 2024–2025 Tax Years

Income Bracket	Tax Rate
Up to €28,000	23%
From €28,001 to €50,000	35%
Above €50,000	43%

Source: https://www.confetra.com/scaglioni-irpef-detrazioni-2024-2025/.

For the 2024–2025 tax years, the IRPEF schedule provides for three income brackets: 23% up to €28,000, 35% from €28,001 to €50,000, and 43% above €50,000 (D.Lgs. No. 216/2023, art. 1; Law No. 207/2024, art. 1, para. 2).

2.1.1. Recent Developments

Over the last 20 years, the Italian IRPEF system has undergone several changes, which can be summarised in the following table:

Table 5. Evolution of IRPEF Tax Brackets and Rates (2007–2026)

Period	Number of Brackets	Tax Rates
2007–2021	5	23%, 27%, 38%, 41%, 43%
2022–2023	4	23%, 25%, 35%, 43%
2024–2025	3	23%, 35%, 43%
2026	3	23%, 33%, 43%

Source: https://www.confetra.com/scaglioni-irpef/.

Note that the 2026 row is included only to indicate the direction taken by the most recent reform, introduced by the 2026 Budget Law (legge di bilancio 2026).

The point that matters most for this analysis is that, in none of these periods did IRPEF have a general mechanism for automatically indexing tax thresholds to inflation. Every update was discretionary, with no built-in rule for adjusting brackets to prices. During periods of high inflation, such as the post-pandemic years 2021 to 2023, this had a specific consequence: taxpayers whose nominal income rose purely because of inflation were automatically pushed into a higher marginal tax bracket on part of their income, despite seeing no real improvement in their income at all.

2.2. Inflation and Income

Having examined the phenomenon in Italy from a theoretical perspective, attention now shifts to the empirical dimension in order to provide descriptive evidence on the macroeconomic context in which fiscal drag may arise. Household disposable income is not equivalent to the IRPEF tax base. It is used here only as a macroeconomic indicator of households' resources, not as a direct measure of taxable income.

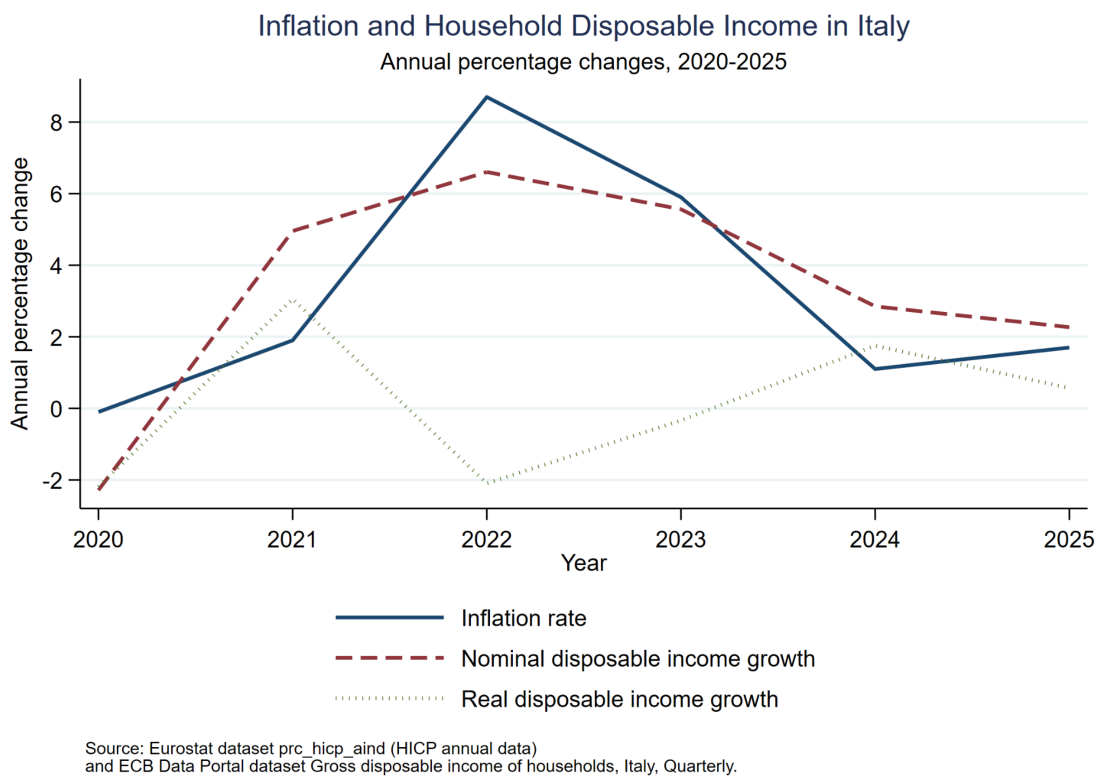

Figure 2. Inflation and Household Disposable Income in Italy. Inflation rate (solid line), nominal disposable income growth (dashed line) and real disposable income growth (dotted line), expressed as annual percentage changes over 2020–2025. Source: Eurostat (prc_hicp_aind) and ECB Data Portal (gross disposable income of households, Italy).

The graph, entitled *"Inflation and Household Disposable Income in Italy"*, compares three variables: the inflation rate, nominal growth, and real growth in disposable income. The inflation data are obtained from Eurostat's *prc_hicp_aind* dataset, filtered by year and country, while disposable income data are taken from the ECB dataset on the gross disposable income of Italian households. Since the latter reported data only quarterly, and gross disposable income is a flow variable, the values were aggregated annually by simply summing them. Nominal growth in disposable income was subsequently calculated and, by subtracting inflation, real growth was obtained. This gives a good approximation, but real growth was calculated by deflating nominal growth according to the formula:

$$g_t^{real} = \frac{1 + g^{nominal}}{1 + \pi} - 1$$

A close examination of the curves shows that, in 2022, the inflation rate exceeded the nominal growth of disposable income, generating a negative real variation. This means that incomes increased in monetary terms, but not sufficiently to offset the rise in prices. In 2023, there was a limited recovery, while it was necessary to wait until 2024, when the reduction in inflation allowed for a recovery in real disposable income.

This analysis is fundamental because it shows a possible divergence between nominal and real income dynamics. This divergence provides the basis for the subsequent analysis, namely to verify whether, during the same period, nominal income growth was also accompanied by an increase in the tax burden.

2.3. Effects on the Tax Burden

Section 2.2 showed that nominal and real income parted ways during the inflationary years: incomes rose in money terms, but in 2022 prices rose faster, so real disposable income fell before recovering later. That is a relationship between two macroeconomic series, and it stops short of the question this thesis sets out to answer. The fact that incomes lagged behind prices does not by itself tell us that the tax system took a larger share of those incomes, nor why. A taxpayer whose nominal income rises can pay a higher average rate for two very different reasons: because Parliament chose to raise it, or because the brackets stood still while inflation carried income across them. Telling the two apart calls for a counterfactual, the tax actually paid set beside the tax that would have been paid had the IRPEF schedule kept up with inflation.

The approach follows the microsimulation literature on bracket creep *(Immervoll, 2005; Balladares and García-Miralles, 2024; Sanz Sanz and Arrazola Vacas, 2025)*, applied to the administrative data published by the Italian Ministry of Economy and Finance. For each tax year the Department of Finance releases the full population of personal income tax returns, grouped into cells by income class and age class, with every cell reporting the number of taxpayers, total taxable income, gross tax, deductions and net tax. The analysis draws on these data for the tax years 2020 to 2024. Prices are measured with the ISTAT NIC index rather than the Eurostat HICP used in Section 2.2. The choice is deliberate: in the absence of a general IRPEF indexation rule, the NIC provides a plausible domestic benchmark for revaluing tax thresholds in the counterfactual exercise. An alternative would be the FOI index, commonly used in other Italian indexation mechanisms, so the estimates should be read as conditional on this methodological choice.

The exercise runs cell by cell. Applying the schedule to the cell mean, rather than to each return separately, introduces an approximation. Since the schedule is convex, the tax computed on average income can never exceed the average of the taxes owed within the cell, and the two are equal when the cell sits inside a single bracket. The two diverge only when a cell straddles a threshold. In the middle of the distribution, where fiscal drag bites, the IRPEF classes are narrow enough that this bias remains minor, a claim the validation below makes precise: where the reconstructed gross tax tracks the recorded figure to within roughly one per cent, the aggregation term is no larger. For each income cell in year *t*, we recompute the tax by applying the statutory bracket schedule to the cell's average taxable income, then divide that gross statutory tax by taxable income to get the effective average rate. We run this twice. The first pass uses the brackets actually in force in year *t*. The second uses a counterfactual schedule: the 2020 brackets scaled up by cumulative inflation and otherwise unchanged. The year 2020 is taken as the base because it is the last full tax year before the post-pandemic inflation, and therefore the last point at which the IRPEF schedule and the price level were still aligned; anchoring the indexed counterfactual there lets the cumulative measure track the entire drift that opened up once inflation accelerated from 2021. The choice is one of timing, not of distributional convenience. 2020 was itself an anomalous year, with incomes shaped by the pandemic and by emergency support, but the base year fixes only the schedule to be revalued: the income distribution is read from each year's own data, so the anomaly does not propagate into the later figures. Both rates come from the same statutory formula rather than from the net tax recorded in the data, which keeps them strictly comparable, since the only thing that moves between them is where the thresholds sit. The fiscal drag of year *t* is the gap between the two:

$$FD_t = ETR_{\text{actual}}(Y_t) - ETR_{\text{indexed}}(Y_t)$$

Here ETR is the effective average tax rate and Y_t is taxable income in year *t*. Because Y_t is the same in both terms, the comparison holds income fixed and lets only the schedule vary, which is what isolates the effect of indexation. A positive gap means taxpayers paid a higher average rate than a fully indexed schedule would have charged on the same income. That is fiscal drag. The aggregate figure reported below is the cell-level result weighted by the number of taxpayers, which equals the extra revenue from non-indexation divided by total taxable income. Two assumptions are built into this design and should be made explicit. It assumes no labour-supply response: taxpayers do not change hours or participation when bracket creep raises their marginal and average rates. And it assumes no general-equilibrium feedback: gross wages and prices are taken as observed, so wage settlements that pass inflation through to nominal pay are left outside the model. This is the standard *"morning-after"* frame of the microsimulation literature *(Immervoll, 2005; Balladares and García-Miralles, 2024)*. The estimates below should therefore be read as the impact effect: a behavioural contraction in labour supply would reduce the extra revenue that non-indexation generates, so the euro figures in particular are an upper bound on what the Treasury actually retains.

One question has to be settled before the results can be trusted: does the bracket model reproduce the tax that taxpayers actually paid? For most of the distribution it does. Across the income cells that account for the bulk of revenue, the gross tax computed from the statutory schedule sits within roughly one per cent of the gross tax recorded in the administrative data. The gap widens only at the bottom, in the lowest income classes, where the no-tax area and the tax credits for employment and pension income pull effective taxation below the statutory rate (Balladares and García-Miralles, 2024). The model does not reproduce those credits, so it overstates tax there. This is a familiar limitation of any schedule-based exercise, and it leaves the middle and upper-middle classes untouched.

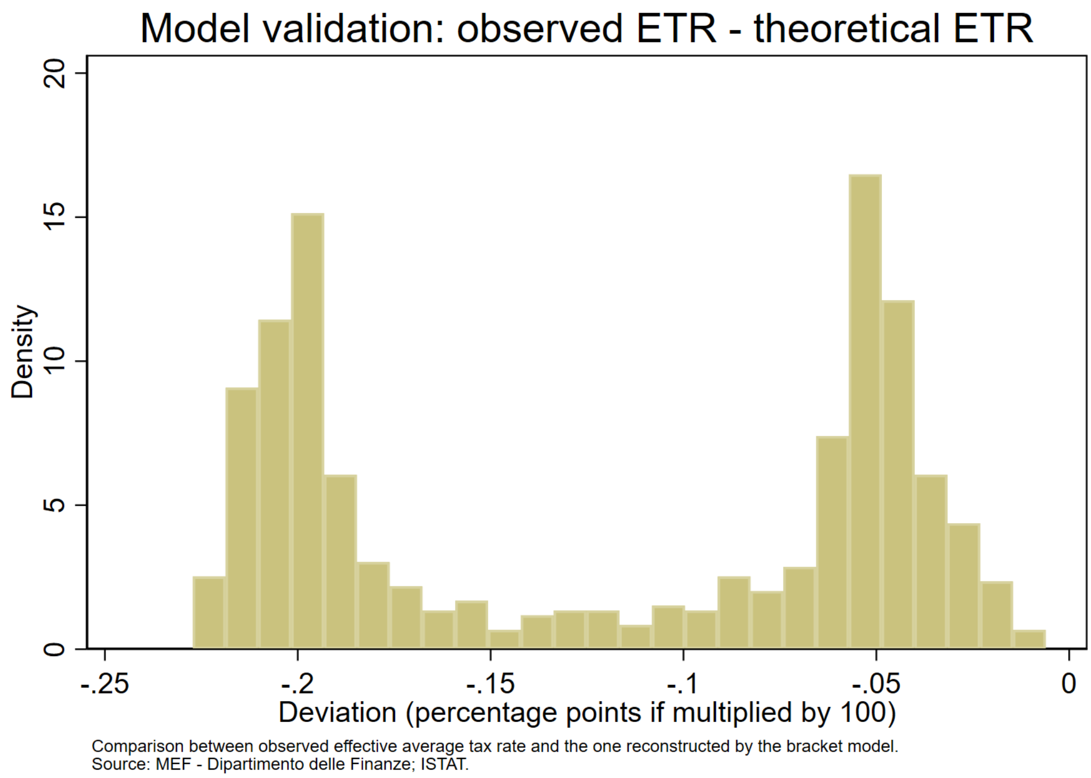

Figure 3. Model validation: distribution of the gap between the observed effective average tax rate and the rate reconstructed from the statutory schedule, tax years 2020–2024. Source: MEF – Dipartimento delle Finanze; ISTAT.

2.3.1. The magnitude of fiscal drag, 2020–2024

Table 2.3 gives the aggregate fiscal drag for each year, in percentage points of the effective average tax rate. It reports two figures. The *cumulative* measure compares each year with the indexed 2020 schedule, so it picks up the whole distance between the tax actually paid and a world in which the brackets had been indexed from the start of the period. The *marginal* measure compares each year only with the one before it, so it picks up the drag produced by the inflation of that single year. Because the marginal measure compares each year only with the one before, it does not depend on the choice of base year at all; the base-year decision affects only the level of the cumulative series, which is anchored to 2020 by construction. The distinction between the two measures matters here because two of the five years contain a tax reform, which the marginal and cumulative figures pick up differently.

Table 6. Aggregate Fiscal Drag in Italy, 2020–2024 (percentage points)

Tax year	Marginal	Cumulative
2020	0	0
2021	0.102	0.102
2022	-0.251	-0.164
2023	0.275	0.077
2024	-0.448	-0.373

Source: own elaboration on MEF – Dipartimento delle Finanze and ISTAT data.

The pattern in the table would not show up in the descriptive series of Section 2.2. In 2021 the five-bracket schedule inherited from the previous decade was still in force, inflation was modest at 1.9 per cent, and both measures record a small positive drag of about a tenth of a point. This is bracket creep with nothing else mixed in: incomes drifted up against fixed thresholds and the average rate rose, with no legislative act behind it.

In 2022 and 2024 the sign turns negative. This does not mean that inflation stopped pushing incomes across thresholds, since 2022 saw the sharpest price rise in forty years; it means that each of these years carried a reform. Rather, discretionary tax cuts in those years more than offset positive bracket creep, making the net measured effect negative. Pure creep, by construction, remains non-negative (see Table 2.4). In 2022 the five brackets became four and the second-bracket rate fell from 27 to 25 per cent. In 2024 four became three and the second bracket was folded into the first. These cuts gave back more than inflation had taken, so the effective average rate fell on net. The counterfactual records this rather than concealing it: a negative value means that discretionary policy returned more to taxpayers in that year than bracket creep removed.

The negative signs in 2022 and 2024 deserve a closer look, because they combine two forces that pull in opposite directions. Table 2.4 separates them by decomposing the cumulative figure of Table 2.3 into two parts. The first is pure bracket creep, the effect of non-indexation alone, obtained by comparing each year's schedule with the same schedule whose thresholds are scaled by cumulative inflation. The second is the residual attributable to the discretionary reforms. The first component is non-negative by construction, since holding the structure of the schedule fixed, indexing the thresholds can only lower the average rate. The decomposition shows that bracket creep was positive and rising throughout the period, reaching about nine tenths of a point by 2024. The negative net values therefore arise entirely from reforms that returned more than creep had removed.

Table 7. Decomposition of Cumulative Fiscal Drag: Pure Bracket Creep and Reform/Residual Component, 2020–2024

Tax year	**Pure creep (≥ 0)**	**Reform effect**	**Net (= Table 6, cumulative)**	**Pure creep (€ bn)**
2020	0	0	0	0
2021	+0.102	0	+0.102	~ 0.8
2022	+0.526	-0.684	-0.164	~ 4.5
2023	+0.788	-0.713	+0.077	~ 7.3
2024	+0.878	-1.253	-0.373	~ 8.5

Source: own elaboration on MEF – Department of Finance and ISTAT data. The “pure creep” column coincides with the “no indexation” column of Table 9. The reform/residual effect is obtained by difference between the net cumulative effect and pure bracket creep. Pure creep revenue is computed as $Pure\ creep\ revenue_t = (Pure\ creep_t/100) \times Aggr.\ taxable\ income_t$.

The table separates the two forces. Pure bracket creep is positive and rising throughout, reaching about nine tenths of a point by 2024, while the reform effect runs the other way and grows as the 2022 and 2024 cuts accumulate against the frozen 2020 structure; the net is their sum, which is why it turns negative only in the reform years. The 2021 row, where the reform effect is zero and the net equals pure creep, confirms the decomposition is consistent.

Figure 2.3 places the two measures side by side. The marginal line swings more widely because it looks back only one year; the cumulative line is steadier because it averages the whole path from 2020. In the reform years neither line isolates fiscal drag on its own, since the reform and the inflation are bound together in the same figure. The cleaner reading comes from the two years without a reform. In 2021 and 2023 the marginal values are positive, and they are larger in 2023, where inflation ran at 5.6 per cent, than in 2021, where it ran at 1.9. The intensity of the drag tracks the intensity of inflation once discretionary reform is set aside.

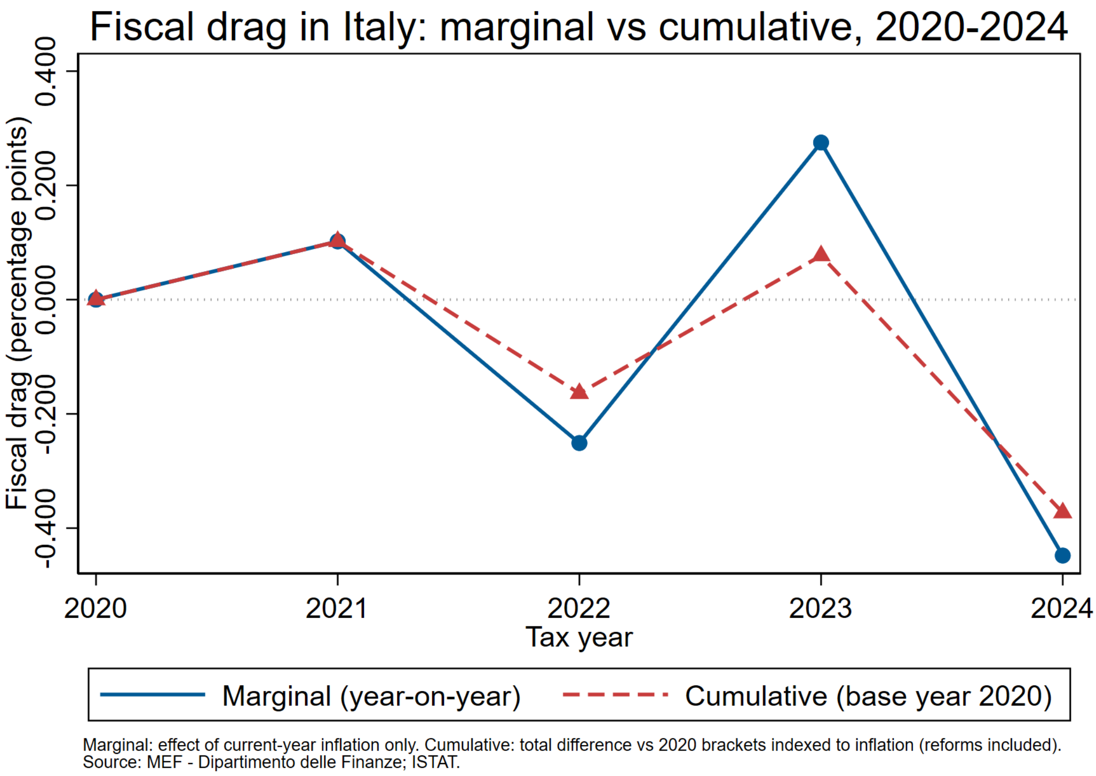

Figure 4. Fiscal drag in Italy, marginal versus cumulative measure, 2020–2024. The marginal series captures current-year inflation only; the cumulative series captures the total gap relative to the 2020 schedule indexed to inflation, reforms included. Source: MEF – Dipartimento delle Finanze; ISTAT.

2.3.2. The distribution of fiscal drag across income classes

An aggregate number hides who actually pays. Fiscal drag works through the crossing of thresholds, so its weight falls unevenly along the distribution. It is heaviest where taxpayers bunch around a bracket boundary and the marginal rate jumps. It is weak at the bottom, where the no-tax area cancels it, and weak at the very top, where income already sits in the highest bracket and a shift in the lower thresholds changes little. Figure 2.4 breaks the cumulative fiscal drag of 2024 down by income class.

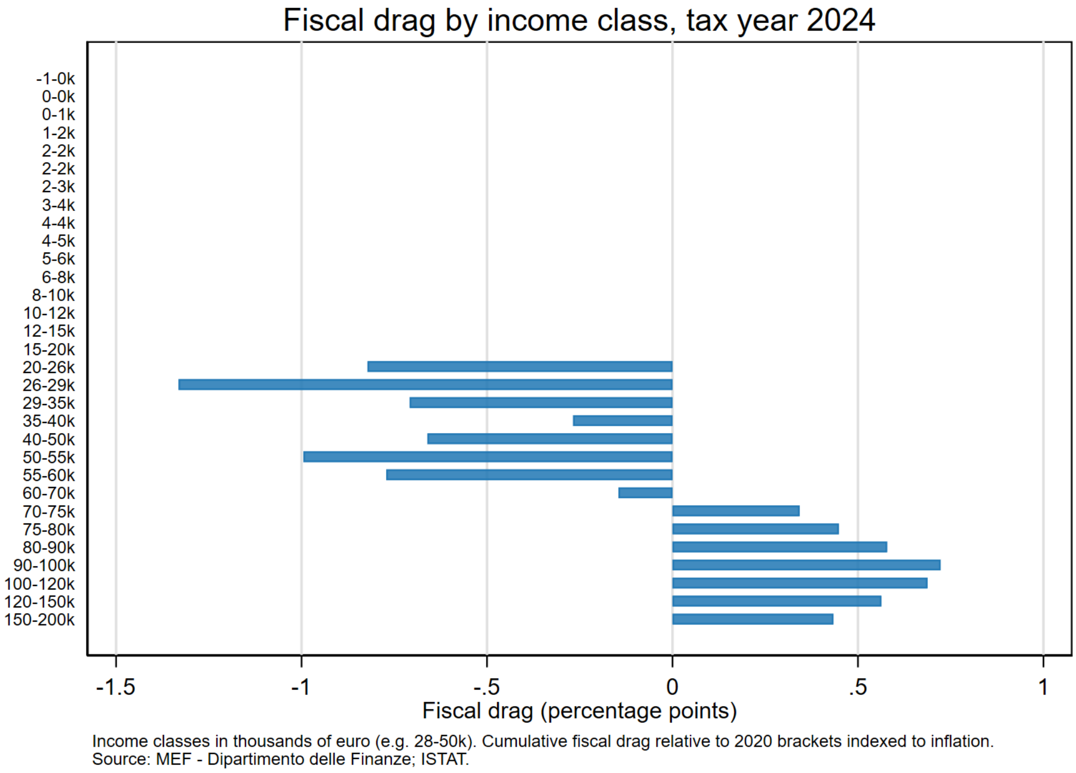

Figure 5. Fiscal drag by income class, tax year 2024 (cumulative measure relative to the 2020 schedule indexed to inflation). Income classes in thousands of euro. Source: MEF – Dipartimento delle Finanze; ISTAT.

The breakdown confirms that the weight is not shared evenly. The classes in the middle of the distribution show the strongest exposure, which is what Chapter 1 led us to expect and what Heer and Süssmuth (2013), Balladares and García-Miralles (2024) and García-Miralles et al. (2025) find for other European systems. These are the households that the no-tax area does not protect and that maximal rates do not yet apply to, and they are the ones that feel the silent transfer when nominal income drifts up against fixed thresholds. The 2024 reform pulls several classes below zero, because lower rates offset the accumulated drag, but the structural fact stands: the middle of the distribution is where the exposure sits.

2.4. Interpretation

Sections 2.2 and 2.3 work on two levels. The descriptive level recorded the split between nominal and real income during the inflation years and the rough co-movement of income and IRPEF revenue. The analytical level built an inflation-indexed counterfactual, measured the part of the tax burden that comes from leaving the schedule unindexed, and showed where along the distribution it falls. Put together, they support a reading that neither could give on its own. Inflation eroded real incomes, and the progressive structure of the IRPEF turned part of that erosion into a higher effective tax burden wherever a reform did not step in first.

2.4.1. Reduction in Real Income

The first conclusion is about the erosion of real income. Section 2.2 already showed it; Section 2.3 gives it a number. Nominal disposable income grew over 2021–2024, but households were not correspondingly better off, and in 2022, the peak inflation year, higher prices wiped out the nominal gains and then some. That split is what makes fiscal drag possible. When income rises in nominal terms against a schedule that nobody has adjusted, the state's share can rise even though the taxpayer has gained nothing real. The counterfactual makes the mechanism concrete. Without the 2022 and 2024 reforms, the positive marginal drag of the non-reform years would have accumulated year after year, lifting the effective average rate through nothing more than inflation meeting progressivity. The reforms broke that accumulation. They did so through one-off legislative choices rather than any standing rule, which is the thread Chapter 3 picks up.

2.4.2. Impact on the Middle Classes

The second conclusion is about who carries the burden. Guzzardi, Palagi, Roventini and Santoro (2022) find that the loss of real income from inflation did not fall evenly in Italy: the hardest hit were lower-income groups and the young, while middle-income groups saw a smaller contraction. That does not push the middle classes to the margin of a study of fiscal drag, for two reasons that the decomposition in Section 2.3.2 makes plain. First, middle-income households pay a large share of total IRPEF, so even a modest rise in their effective rate adds up to a substantial sum in the aggregate. Second, they sit exactly where bracket creep bites hardest, bunched around the thresholds where the marginal rate steps up, outside the reach of the no-tax area and below the top bracket *(Balladares and García-Miralles, 2024)*. Fiscal drag therefore presses hardest on the centre of the distribution, where a rise in nominal income with no real gain behind it can still raise the tax bill.

A caution belongs at the end, in keeping with the partly descriptive nature of the evidence. The administrative data come in income and age cells, and the counterfactual rests on the statutory schedule rather than on the full set of deductions and credits, so the exercise approximates the fiscal drag borne by any single taxpayer rather than pinning it down exactly. The order of magnitude is consistent with independent work. *The Osservatorio sui Conti Pubblici Italiani* put the additional IRPEF revenue from fiscal drag on dependent employment at roughly €0.3 billion in 2024. That figure is gross, covers a single year and a single category, and is not directly comparable with the cumulative, net-of-reform estimates in Table 2.3. It still points the same way: even in a reform year, the underlying bracket creep was positive, as the pure-creep column of Table 2.4 also shows. What the analysis settles is the structural point, not a precise euro figure. In a progressive system with no automatic indexation, inflation raises the effective tax burden without any visible policy step. That rise lands with particular weight on the middle of the income distribution, and only discretionary legislation, which by its nature comes and goes, keeps the burden from building up year on year.

3. A European Comparison of Fiscal Drag

Having analysed the relationship between inflation, disposable income, and IRPEF revenue in the Italian context, this chapter widens the perspective to other European tax systems. As discussed in Section 1.3.2, fiscal drag does not arise solely from the combination of a progressive tax system and high inflation. Its intensity also depends on the rules that govern how thresholds, deductions, tax credits, and transfers are updated over time.

The comparison focuses on Italy, Germany, and Sweden, three countries that handle the erosion of real tax thresholds in different ways. In Italy, the main changes to personal taxation occur only through discretionary legislative reform. Germany relies instead on recurring adjustments to its thresholds and tax-free allowance, intended to limit the effects of the so-called "*Kalte Progression*". Sweden indexes the threshold of its state income tax to a fixed rule, though the legislator may still suspend or limit that adjustment. These three approaches define different degrees of protection for taxpayers when nominal incomes rise without a matching gain in purchasing power, and the aim of the chapter is to show how the design of a tax system shapes its exposure to fiscal drag rather than to measure that exposure directly. The chapter proceeds in four steps. Section 3.1 sets out the comparative method and the criteria used to select the countries. Section 3.2 describes how each country adjusts its tax thresholds. Section 3.3 draws on the OECD's *Taxing Wages 2026* to compare the taxation of labour and its recent changes. Section 3.4 turns the comparison back on Italy, where the absence of a general and automatic indexation mechanism leaves taxpayers more exposed.

3.1. Methodology and Country Selection

The comparison relies mainly on the OECD's *Taxing Wages 2026*. The report compares labour taxation across countries using standardised indicators calculated for fixed worker and household types. The main indicator is the tax wedge, which is the gap between the total cost of a worker to the employer and the worker's net take-home pay. It includes personal income tax, the social security contributions paid by the employee and the employer, and any cash transfers the household receives. Because the tax wedge combines income tax with contributions and transfers, it cannot be read as a direct measure of the Italian IRPEF or of the income taxes of the other countries. It remains useful here because it shows the total burden on labour, and how that burden moves when thresholds, rates, contributions or transfers are changed. The analysis uses a single worker without children, earning the national average wage. The OECD calculates this profile for every country, so using it removes the variation that comes from household composition and child-related transfers. The cost of this choice is that the comparison covers only one point in the income distribution and says nothing about the redistributive effect of each system as a whole. It describes the tax treatment of average employment income only. For Italy, *Taxing Wages* uses a backward-looking approach: the figures labelled 2025 refer to the rules in force in the 2024 fiscal year. This chapter therefore uses the same tax year as the analysis in

Chapter 2. The three countries were chosen for the way each updates its tax parameters against inflation. Italy adjusts them only through discretionary reform. Germany does so through recurring legislative measures, and Sweden ties them to an indexation rule. This is the difference that matters for fiscal drag, because the way a system updates its thresholds determines how far taxpayers are protected when nominal incomes rise without a real gain.

3.2. Indexation Mechanisms in Italy, Germany and Sweden

As stated previously, the intensity of fiscal drag depends not only on the progressivity of income taxation, but also on the methods through which thresholds, deductions, and other tax parameters are updated over time. The comparison between Italy, Germany, and Sweden specifically concerns the rules governing the updating of the main parameters of income taxation.

3.2.1. Italy: No Automatic General Indexation of IRPEF Thresholds

With regard to Italy, this section will not be explored in great detail, as it has already been thoroughly examined in Chapter 2 of this thesis. Italian IRPEF thresholds are not subject to an automatic indexation mechanism. This is the core issue. Changes to tax brackets, rates, and deductions are introduced through specific legislative measures. Recent reforms have modified the structure of the tax, but have not provided for an automatic annual adjustment of thresholds in line with inflation.

3.2.2. Germany: Regular Legislative Adjustments to Offset Cold Progression

In Germany, fiscal drag is known as Kalte Progression. Like Italy, Germany has no system of automatic indexation for the personal income tax. The legislator, however, adjusts the basic allowance *(Grundfreibetrag)* and the thresholds of the rate schedule fairly regularly. The basic allowance and the bracket thresholds were raised to account for inflation in 2021, and again in 2022, and the allowance has been increased in the years since. These adjustments are not occasional one-off measures, which is what distinguishes the German case from the Italian one. They are a recurring instrument used to contain the effects of progressivity when nominal incomes rise. The adjustments still depend on legislative decisions, so they do not always keep pace with inflation. In 2025, for instance, gross earnings rose by more than the increase in the thresholds, so part of the cold progression was not offset.

3.2.3. Sweden: Rule-Based Annual Indexation and Its Exceptions

Sweden uses a more structured approach. The threshold of the central government income tax is indexed by rule to the consumer price index plus 2% each year. The adjustment is therefore automatic and does not require a fresh legislative decision, which is the main difference from systems where any change to the thresholds depends on the legislator acting. By updating the threshold to prices every year, the Swedish rule reduces the risk that inflation raises the tax burden on labour through legislative inaction. The indexation is not absolute. Swedish law allows the increase set by the rule to be limited, and the OECD reports that this has happened in six years: 2004, 2005, 2006, 2016, 2017, and 2024. Sweden therefore shows that a rule-based mechanism gives steadier protection against fiscal drag than discretionary adjustment, but that it remains open to legislative override.

3.3. Comparative Evidence on Fiscal Drag

This section matters because different updating mechanisms do not automatically produce the same outcome for the tax burden. What also counts is the path of wages, of contributions, of tax credits, and of the reforms adopted in each year.

3.3.1. Differences in Countries' Exposure to Fiscal Drag

To compare the three countries and their exposure to fiscal drag, this section uses the tax wedge reported by the OECD. The tax wedge does not measure the pure effect of fiscal drag; it is used here as an indicator of how the burden on labour changes. It is the share of the total cost of labour that does not reach the worker as net income, because it is absorbed by personal income tax, the employee's social security contributions, the employer's social security contributions, and any payroll taxes. An example makes this clearer. Consider the reference worker: a single person without children, earning the national average wage.

Table 8. Tax Wedge and Annual Change for Single Worker at the Average Wage, 2025

Country	Tax Wedge 2025	Change 2025/2024
Germany	49.3%	+1.34 p.p.
Italy	45.8%	-1.21 p.p.
Sweden	41.1%	-0.37 p.p.

Source: OECD (2026), *Taxing Wages 2026: The Progressivity of Labour Taxation in OECD Countries*, p. 26, Table 1.1.

The table shows that, for every 100 euro of labour cost in Germany, about 49.3 euro is absorbed by taxes and contributions. Italy follows at 45.8 euro, and Sweden at 41.1 euro. The overall levy on labour in 2025 was therefore highest in Germany. The last column gives the change in the tax wedge between 2024 and 2025, in percentage points. It shows whether the levy on labour rose or fell from the previous year. In Germany the change was positive, at +1.34 points, which brought the wedge to 49.3%. In Italy and Sweden it was negative, at −1.21 and −0.37 points respectively. These changes need to be read with caution. The tax wedge does not depend on income tax alone; it also includes the social security contributions of employee and employer, and any tax credits or transfers. An increase or a decrease in the wedge is therefore not a direct measure of fiscal drag, and some care is needed in reading it. The German figure still says something useful. According to the OECD, the +1.34-point increase was driven mainly by higher social security contribution rates for sickness and long-term care insurance and by the end of the inflation compensation premium, but it was also driven in part by gross earnings rising faster than the adjustments to the income tax thresholds, which is fiscal drag. Germany makes regular legislative adjustments against *Kalte Progression*, yet in 2025 those adjustments did not fully offset it. This points to a broader reason for caution in reading the table. The wedge fell in Italy and Sweden and rose only in Germany, but a falling wedge does not mean that fiscal drag was absent. In Italy the decline reflects the 2024 reform, which lowered the IRPEF rates: the same reform that turned the cumulative fiscal drag negative in Chapter 2. The tax wedge captures the combined result of everything that changed in a given year, reforms, contribution rates, credits and wage growth together, so it cannot separate fiscal drag from the rest. Its role in this chapter is to compare how the three systems behave, not to measure the drag itself, which for Italy was already measured directly in Chapter 2. Read in this way, the table carries one clear message. Where a system adjusts its parameters, whether through reform in Italy or through regular correction in Germany, the burden can still move in either direction. The protection each system offers against inflation depends on the rule, or the political decision, behind every adjustment.

3.4. Implications for Italy

The comparison suggests that the Italian problem is the lack of a rule for updating the parameters. The level of the tax burden is not unusual; it sits between the German and the Swedish figures. Italy does correct for inflation. The reforms of 2022 and 2024 lowered the burden and pushed the cumulative fiscal drag below zero. But these corrections came through discretionary reforms, timed by the budget cycle rather than by prices. In 2021 and 2023, when no reform was passed, the rise in the tax burden went uncorrected, and it was largest in 2023. Germany corrects cold progression through frequent legislative adjustment, and Sweden through an indexation rule. In both the correction is more regular than in Italy, even if fiscal drag is never fully removed.

3.4.1. A rule-based mechanism and its simulation

The mechanism proposed here revalues the IRPEF parameters in line with the ISTAT NIC index. The revaluation applies to the bracket thresholds, and also to the no-tax area and the income-related credits: the validation in Section 2.3 showed that these credits, which are not indexed, shape effective taxation at the bottom of the distribution, so a rule that left them out would miss part of the drag *(Balladares and García-Miralles, 2024)*. To keep a fiscal margin, the parameters are revalued only when cumulative inflation since the last revaluation passes a threshold, set here at 3%; this also preserves part of the stabilising role discussed in Section 1.4.3. The legislator keeps the power to suspend the rule by explicit act, as in Sweden, so that any rise in the burden becomes a deliberate decision. The microsimulation of Chapter 2 makes it possible to test the rule. The counterfactual built there is itself the simulation of full indexation, so the proposal can be assessed by comparing three updating rules on the same 2020 bracket structure and the same income distribution: no indexation, full indexation, and the triggered rule. The "no indexation" column reproduces the pure bracket creep of Section 2.3.1 (Table

2.4): unlike Table 2.3, which tracks the burden as it actually evolved with reforms included, Table 3.2 holds the 2020 structure fixed and varies only the updating rule, isolating indexation from discretionary reform. The threshold is passed in 2022 and 2023 but not in 2021 (inflation 1.9%) or 2024 (1.0%), so the rule acts twice in five years. Table 3.2 and Figure 3.1 report the residual drag against full indexation. Without indexation it accumulates to about nine tenths of a point by 2024. Under the triggered rule it is close to zero in 2022 and 2023, around a tenth of a point in 2021, and below one twentieth in 2024. The rule removes most of the drag, including the 2023 erosion that the real schedule left in place, and it does so with only two revaluations.

Table 9. Residual Fiscal Drag under Alternative Indexation Rules, 2020–2024 (p.p., benchmark = full indexation)

Tax year	**No indexation**	**Triggered rule ($\theta = 3\%$)**
2020	0	0
2021	0.102	0.102
2022	0.526	0
2023	0.788	0
2024	0.878	0.047

Source: MEF – Dipartimento delle Finanze, ISTAT.

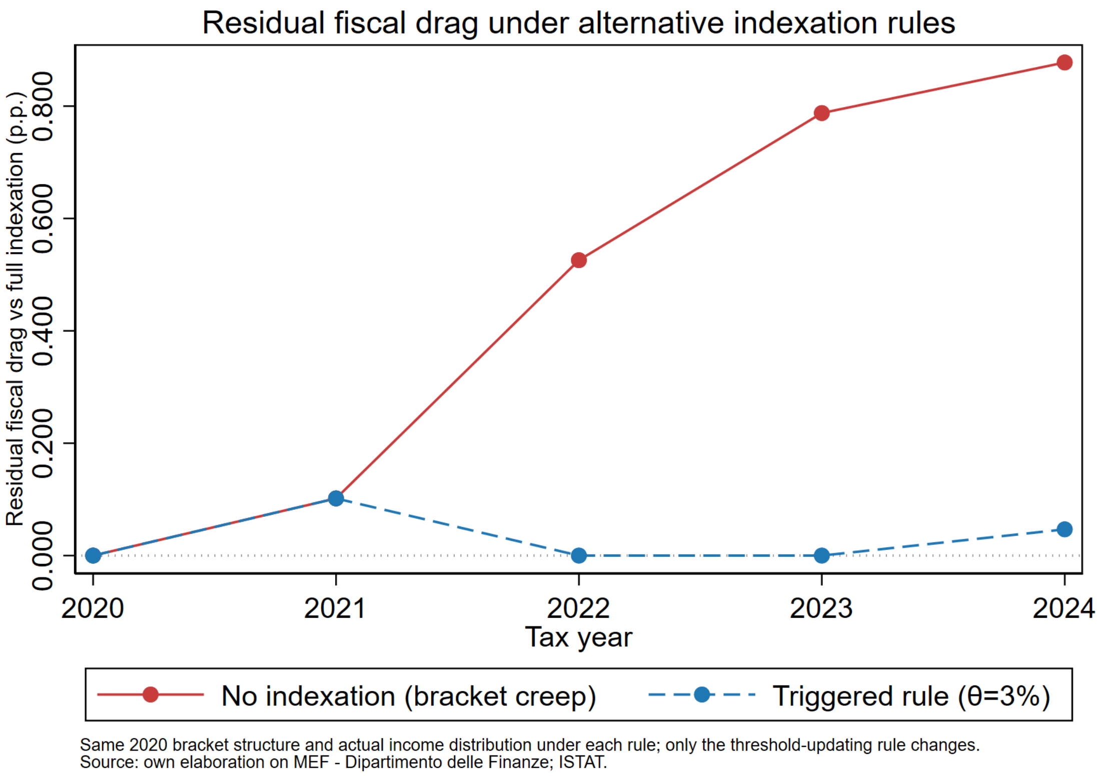

Figure 6. Residual fiscal drag under alternative indexation rules, 2020–2024. Each rule applies the same 2020 bracket structure to the actual income distribution; only the updating rule changes. Source: own elaboration on MEF – Dipartimento delle Finanze; ISTAT.

Neither version is free of cost. Full indexation gives complete protection, but it forgoes the real revenue that non-indexation generates and removes the stabiliser of Section 1.4.3. The triggered rule keeps part of that revenue and leaves only small drift uncorrected. The choice between them is political, and the data cannot settle it. The simulation also inherits the limits of the analysis behind it: the data are grouped into cells, the model uses the statutory schedule, and the 3% threshold is illustrative. These do not change the main result. A rule tied to inflation, even a partial one, would have removed most of the drag of 2020–2024 with two adjustments, while the actual system corrected it unevenly.

4. Conclusions

4.1. Synthesis of the results

This thesis set out to measure fiscal drag in the Italian IRPEF and to place it in a European context. The theoretical chapter established the mechanism: in a progressive system whose parameters are not indexed, inflation lifts nominal incomes against fixed thresholds and raises the effective average tax rate even when real income has not changed. The empirical chapter measured that mechanism for Italy over the tax years 2020 to 2024, using the administrative micro-data of the Department of Finance and an inflation-indexed counterfactual; it separated the drag of a single year from the drag accumulated since 2020 and traced the burden across income classes. What the data show turns on whether the legislator intervened. In 2021 and 2023, the years without a reform, there is a positive fiscal drag that grows with inflation, the clearest form of bracket creep. In 2022 and 2024, discretionary rate cuts more than offset the drag and turned the measured effect negative; Italy does correct for inflation, but through irregular intervention rather than a standing rule. The comparative chapter set these results against Germany and Sweden. Germany contains cold progression through frequent legislative adjustment; Sweden indexes its central-government threshold to a rule, subject to occasional suspension. The OECD evidence confirms that fiscal drag operates across systems, and that even regular correction does not always offset it. Taken together, the three cases locate Italy's problem not in the level of the tax burden but in the absence of a predictable mechanism for keeping the parameters aligned with prices.

4.2. Critical evaluation

The findings should be read within their limits. The estimate rests on aggregated cells and on the statutory schedule, so it captures the structure of fiscal drag rather than the exact amount borne by any individual taxpayer, and it understates the effect at the bottom of the distribution, where unmodelled credits matter most. The counterfactual also abstracts from behaviour: it holds the income distribution fixed and models no labour-supply or general-equilibrium response, so it measures the mechanical impact of fiscal drag rather than its net effect in equilibrium. Its cumulative measure is anchored to the 2020 schedule, while the year-on-year measure, on which the link between drag and inflation rests, does not depend on that base. Its order of magnitude is consistent with the independent estimate of the *Osservatorio sui Conti Pubblici Italiani*, which lends it some external support. The comparative analysis relies on the OECD tax wedge, a broad indicator that bundles income tax with contributions and transfers, and therefore cannot isolate fiscal drag on its own; it compares the behaviour of the three systems but does not measure the drag, which for Italy was done in Chapter 2. The policy simulation inherits these limitations and adds one of its own, the choice of activation threshold, which is illustrative. None of these caveats overturns the central finding. Each marks the boundary within which it holds.

4.3. Possible interventions

The analysis points to one broad recommendation: replace discretionary correction with a rule. The mechanism proposed in Section 3.4 indexes the IRPEF thresholds, the no-tax area, and the income-related credits to the NIC index, revaluing them once cumulative inflation passes a set threshold and leaving the legislator an explicit power to suspend the adjustment. Applied to the 2020–2024 data, the simulation shows that such a rule would have neutralised most of the period's fiscal drag, and in particular the 2023 erosion that the real system left uncorrected, while acting only twice and preserving a fiscal margin in the low-inflation years.

The choice between full and partial indexation remains open, and it is properly a political one. Full indexation protects taxpayers completely but forgoes revenue and removes a stabiliser;

a triggered rule shields them from the large erosions while keeping some fiscal room. What the evidence of this thesis supports is the narrower, firmer claim that some rule is better than none. A rule tied to inflation also makes the rising burden visible: rather than accumulating quietly until a reform, timed by the budget cycle, happens to reverse it, the increase has to be argued for in the open. That visibility, as much as the revenue indexation returns, is what makes the case for it.

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